Nova

Reimagining the Academic Interface

Document Type

Software Requirements Specification (SRS)

Project Timeline

April 1, 2026 – May 20, 2026

Prepared By

Gargee Kakaty & Kaustuva Kashyap

Live Deployment

https://gargee1989.github.io/Nova/

Table of Contents

Contents

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document details the functional and non-functional requirements for **Nova**, a GUI-centric mock university website. The primary purpose of this project is to visually simulate a real university portal by emphasising layout structure, visual hierarchy, consistency, and intuitive navigation.

1.2 Document Conventions

This document follows the standard IEEE formatting guidelines for an SRS. Requirements are classified into Functional Dependencies (FR) and Non-Functional Dependencies (NFR), strictly focusing on frontend GUI principles and browser-based logic rather than backend functionality.

1.3 Project Scope

Nova encompasses a multi-page mock university portal with 15+ interconnected pages spanning academics, admissions, campus life, alumni, faculty, scholarships, and more. It acts as a practical sandbox for experimenting with UI/UX design and browser-based logic, communicating institutional information through thoughtful visual design. The project completely avoids external frameworks, relying purely on core web technologies (HTML5, CSS3, JavaScript).

2. Overall Description

2.1 Product Perspective

Nova is a standalone frontend prototype. It addresses the observation that many academic websites focus solely on functional information delivery and fail to convey the essence and atmosphere of the university. The product allows users to intuitively feel the spirit, academic ambition, and campus life of the institution through thoughtful visual design.

2.2 Operating Environment

- **Platform:** Standard modern web browsers (Chrome, Firefox, Safari, Edge).
- **Device Support:** Desktop, tablet, and mobile devices (Responsive Design).
- **Deployment:** Hosted via GitHub Pages — no build tools or server required.

2.3 Design and Implementation Constraints

The system must be built entirely using core web technologies without any external CSS or JavaScript frameworks. The user interface must comply with recognised Web Content Accessibility Guidelines (WCAG 2.1) and standard usability heuristics (e.g. Nielsen's 10 Usability Heuristics).

2.4 Key Features

- **Multi-page architecture** — 15+ interconnected pages spanning academics, admissions, campus life, and more.
- **Responsive design** — Consistent experience across desktop and mobile screen sizes.
- **Dynamic navigation** — Dropdown menus, hamburger toggle for mobile, and smooth in-page anchoring.

- **Apply Now flow** — Dedicated student application page.
- **Animated statistics counter** — Live-counting enrollment figures on the homepage.
- **Marquee announcement bar** — Scrolling highlights reel.

3. Tech Stack

No external frameworks or libraries were used. Everything is written in pure, native web technology.

Technology	Role
HTML5	Semantic structure — headers, navbars, cards, forms, sections
CSS3	Styling via Flexbox & Grid; animations, hover states, responsive layouts
JavaScript	Interactivity — menu toggling, counters, form validation, dynamic rendering

4. Functional Dependencies (Requirements)

The following functional dependencies define the interactive behaviours required for the client-side of the Nova platform.

- **FR-01: Navigation Menu Toggling** — The system shall provide an interactive, togglable navigation menu — including a hamburger toggle for mobile — allowing users to seamlessly transition between primary user flows such as browsing courses, admissions, and contact information.
- **FR-02: Interactive Visual Feedback** — All clickable UI elements (buttons, links, cards) must implement interactive hover states and active state styles to provide immediate user feedback upon interaction.
- **FR-03: Client-Side Form Validation** — The system shall implement JavaScript-based form validation to provide immediate feedback on user inputs (e.g. for the Apply Now and contact forms) prior to any mock submission events.
- **FR-04: Dynamic Content Rendering** — The system shall utilise JavaScript to dynamically render content elements — including an animated statistics counter on the homepage and a marquee announcement bar — without requiring full page reloads, simulating modern single-page application behaviours natively.
- **FR-05: Apply Now Flow** — The system shall provide a dedicated multi-field application page (`applynow.html`) enabling prospective students to submit mock admission applications with client-side validation feedback.
- **FR-06: Multi-Page Navigation** — The system shall support seamless navigation across all 17 pages via consistent header navigation and footer links, with smooth in-page anchoring where applicable.

5. Non-Functional Dependencies

- **NFR-01: Responsive Layout** — The GUI must employ CSS3 Flexbox and Grid layouts to ensure the interface is fully responsive, maintaining a balanced design across desktop, tablet, and mobile viewing environments.
- **NFR-02: Semantic Structural Integrity** — The markup must utilise semantic HTML5 tags (e.g. `<header>`, `<nav>`, `<section>`, `<footer>`) to ensure proper document structure for web accessibility and logical organisation.
- **NFR-03: Usability and Aesthetics** — The design must rigorously apply core GUI principles including alignment, contrast, spacing, and consistency. Colour schemes and typography must be emotionally resonant and suitable for an educational platform.
- **NFR-04: Accessibility Compliance** — The interface must comply with Web Content Accessibility Guidelines (WCAG 2.1) and Nielsen's 10 Usability Heuristics.
- **NFR-05: Zero External Dependencies** — The project must not depend on any external CSS or JavaScript frameworks or libraries. All functionality must be implemented in vanilla HTML5, CSS3, and JavaScript.

6. Pages Overview

Nova comprises 17 interconnected pages organised across the following sections:

Page	Path	Description
Home	`index.html`	Landing page with hero, stats, programs, campus life, faculty, testimonials, and events
About	`html/pages/about.html`	University history, mission, and values
Faculty	`html/pages/faculty.html`	Faculty listings and profiles
Events	`html/pages/events.html`	Upcoming university events
Campus Life	`html/pages/campus.html`	Student life, arts, culture, and athletics
Alumni	`html/pages/alumni.html`	Alumni network and stories
Academic Areas	`html/academic/academic-areas.html`	Overview of colleges and departments
Acad. Calendar	`html/academic/calendar.html`	Semester schedule and key dates
Admission	`html/admission.html`	Requirements, process, and tuition fees
All Programs	`html/programs/allprog.html`	Full list of academic programs
Software Dev	`html/programs/sd.html`	B.Sc Software Development detail page
Creative Writing	`html/programs/cw.html`	M.A. Creative Writing detail page
Architecture	`html/programs/arch.html`	B.A. Architecture Design detail page
Cinematography	`html/programs/cine.html`	B.F.A. Cinematography detail page
Scholarships	`html/studentservices/scholarship.html`	Financial aid and scholarship information
Contact	`html/Contact.html`	Contact form and university details
Apply Now	`html/applynow.html`	Student application page

7. Team Breakdown & Roles

The project is collaboratively developed by students of the 4th Semester Integrated Masters in Software Development.

Name / ID	Academic Status	Project Roles & Responsibilities
Gargee Kakaty (IS-241-844-0013)	IMSc Software Dev (4th Sem)	**Co-Developer & Scripter** JavaScript Logic, Form Validation, DOM Manipulation, Dynamic Rendering, and Mobile Compatibility.
Kaustuva Kashyap (IS-241-844-0022)	IMSc Software Dev (4th Sem)	**Co-Developer & Designer** UI/UX Design, CSS3 Layouts (Flexbox/Grid), Wireframing, and Visual Identity.

8. Project Phases & Timeline

The development cycle ran from April 1, 2026 through May 20, 2026, following a GUI-first methodology.

Phase	Focus Area	Dates (2026)	Deliverables
Phase 1	GUI Design & Wireframing	Apr 1 – Apr 10	Identify primary user flows. Create layout wireframes, select typography, and define the educational colour scheme.
Phase 2	Core HTML/CSS Implementation	Apr 11 – Apr 25	Develop semantic HTML5 structure. Apply CSS3 Flexbox and Grid to build responsive layouts for headers, navbars, and cards.
Phase 3	JavaScript Interactivity	Apr 26 – May 10	Implement client-side logic: navigation toggling, hover states, animated counters, marquee bar, and form validation.
Phase 4	Testing & Delivery	May 11 – May 20	Ensure consistent responsiveness across devices. Validate against WCAG 2.1 and Nielsen's heuristics. Final submission and GitHub Pages deployment.